Dynamic Hybrid Chatbots:

Learning When to Rely on Intent Recognition or Direct Generation

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Abstract

Conversational customer-service systems traditionally rely on intent-based pipelines to ensure control and reliability, while recent Large Language Models (LLMs) enable direct end-to-end generation with strong generalization capabilities. Each paradigm presents complementary strengths: intent-based systems struggle with linguistic variability, whereas purely generative models may behave inconsistently in complex scenarios.

This work proposes a dynamic hybrid framework that learns, at inference time, whether a query should be handled through direct generation or intent-conditioned generation. **The approach is evaluated on a large-scale customer service dataset containing over 26,872 synthetic dialogues annotated with 27 intents.** The framework combines a BERT-based intent classifier, two T5 generators, and a lightweight logistic regression router trained through self-supervised meta-labeling derived from comparative BLEU scores.

Results demonstrate that the router learns a non-trivial decision boundary, achieves a balanced trade-off between response quality and computational cost, and adapts across intents and query characteristics. The findings confirm that explicit intent recognition remains valuable when applied selectively, and that dynamic routing provides a principled way to reconcile control and flexibility in modern conversational systems.

1. Introduction

Conversational Artificial Intelligence systems are widely used in customer service applications such as virtual assistants and automated support agents. Traditionally, these systems rely on modular intent-based pipelines, where user input is first mapped to a predefined intent before response generation. This design ensures controllability, interpretability, and reliable task execution in structured environments.

The emergence of Large Language Models (LLMs) has introduced an alternative paradigm: direct end-to-end generation. Transformer-based models can generate context-aware responses directly from raw input, implicitly inferring intent without explicit classification. While this approach improves generalization to varied phrasing, it may exhibit unstable behavior in ambiguous or multi-step scenarios.

Recent work (Bolshinsky et al., 2025) demonstrates that neither paradigm is universally superior: direct generation performs well on simple queries, whereas intent-aware approaches remain advantageous in complex cases. Despite this evidence, existing conversational systems typically commit to a single processing strategy and apply it uniformly to all queries.

This work addresses this limitation by proposing a dynamic hybrid framework that determines, at inference time, whether a query should be handled through direct generation or intent-conditioned generation. By learning a query-dependent routing policy, the system aims to reconcile flexibility and control in modern customer service chatbots.

2. Literature Review

Early customer-service conversational systems were predominantly built around explicit intent recognition frameworks, relying on predefined intent taxonomies to enable structured dialogue control and predictable behavior. Transformer-based classifiers such as BERT (Devlin et al., 2019) significantly improved intent recognition accuracy, and such approaches remain widely used in regulated domains including banking (Lajčinová et al., 2024). Universal intent recognition systems have further demonstrated strong performance in controlled environments (Vasquez-Correa et al., 2021).

However, intent-based pipelines assume stable and well-defined intent boundaries. In practice, customer queries may be ambiguous, implicit, or multi-intent, challenging classical classifiers even when using modern language models (Ahmad et al., 2025). Moreover, pipeline architectures suffer from error propagation, where early misclassification degrades downstream response quality (Lin et al., 2021).

The rise of Large Language Models (LLMs) has introduced an alternative end-to-end paradigm. Transformer architectures (Vaswani et al., 2017) trained at scale demonstrate strong few-shot capabilities (Brown et al., 2020) and unified learning paradigms (Tay et al., 2022), enabling direct response generation without explicit intent supervision. Prompt-based reasoning techniques further enhance their implicit task understanding (Wei et al., 2022). Empirical evaluations show that LLMs can approximate intent detection (He & Garner, 2023), though performance may degrade for complex inputs, affecting user satisfaction and reliability (Bodonhelyi et al., 2024).

Bolshinsky et al. (2025) provide a direct empirical comparison between intent-first pipelines and end-to-end generation strategies in customer service settings. Their results show that direct generation performs competitively on simple queries, while intent-aware systems remain advantageous for ambiguous or multi-goal requests. This finding highlights the complementary strengths of both paradigms.

Recent research has explored hybrid architectures combining structured intent signals with generative models, including intent-guided prompting and integrated dialogue representations (Chun et al., 2025). Other approaches incorporate feedback-based self-improvement mechanisms to enhance response quality without discarding intent supervision (Azov et al., 2024). Nevertheless, routing strategies in these systems remain static, typically relying on predefined rules or confidence thresholds.

Across the literature, a consistent trade-off emerges: intent-based systems offer control and interpretability, whereas generative models provide flexibility and generalization. However, no prior work explicitly proposes a learned, query-dependent routing mechanism that dynamically selects between explicit intent recognition and direct generation at inference time. This unresolved gap motivates the dynamic hybrid framework introduced in this study.

3. Research Question, Objectives, and Innovation

3.1 Research Question

How can a Large Language Model dynamically decide whether to rely on explicit intent recognition or to generate a direct response, depending on the complexity and ambiguity of the user's request?

This question addresses a key limitation in existing architectures: the use of a fixed processing strategy despite the heterogeneous nature of customer service requests. The absence of an adaptive decision mechanism limits both robustness and efficiency.

3.2 Project Objectives and Motivation

The objective of this work is to design and evaluate a dynamic hybrid chatbot architecture that integrates both intent-based and end-to-end generative approaches within a unified framework.

The motivation stems from the observation that simple queries are often handled effectively by direct generation, whereas complex or ambiguous requests benefit from structured intent conditioning. An adaptive mechanism may therefore improve both reliability and efficiency.

3.3 Innovation and Contribution

We introduce a lightweight meta-controller that learns to select the most appropriate generation strategy at inference time. Unlike prior hybrid systems relying on static routing rules, the proposed router learns its policy from empirical performance signals derived through self-supervised comparison of generation outputs.

This adaptive routing mechanism constitutes the main contribution of this work. By enabling per-query strategy selection, the framework aims to improve response quality while controlling computational cost in customer service applications.

4. Methodology and System Architecture

4.1 Overall System Design

We propose a two-stage hybrid architecture that integrates intent recognition, dual response generation pipelines, and a learned routing mechanism within a unified framework. Given a user instruction, the system first predicts an intent using a BERT-based classifier (bert-base-uncased). Based on this prediction, a lightweight router determines whether the response should be generated directly from the instruction or through an intent-conditioned generation path.

The architecture therefore consists of three main components: an intent classifier, two T5-based response generators (t5-small), and a logistic regression router operating on uncertainty-related features extracted from the intent distribution. The simple generation path produces responses directly from the raw instruction, while the complex path augments the input with the predicted intent before generation. Routing decisions are

performed at inference time, enabling adaptive strategy selection without requiring manually annotated routing labels.

4.2 Dataset and Task Definition

Experiments are conducted on the Bitext Customer Support LLM Chatbot Training Dataset, which contains 26,872 synthetic customer–agent dialogue pairs annotated with 27 intent categories. Each example includes a user instruction, an intent label, and a reference response.

Although synthetic, the dataset provides a controlled benchmark for evaluating intent-based and generative response strategies under realistic customer service scenarios. The task is framed as conditional response generation, where the objective is to produce an appropriate agent response given a user instruction, optionally conditioned on a predicted intent.

4.3 Data Splitting Strategy

To ensure rigorous evaluation and prevent information leakage, the dataset is partitioned into four non-overlapping subsets serving distinct roles in the experimental pipeline. The full dataset is first divided into a main training portion (70%) and a temporary hold-out portion (30%) using a fixed random seed. The temporary portion is further split evenly into a meta-development set and a final evaluation set (approximately 15% each).

The main training portion is subsequently divided into a training set (90%) and a validation set (10%), resulting in approximately 63% of the full dataset used for model training and 7% for validation. The training and validation splits are used exclusively for learning the intent classifier and generation models.

The meta-development set is reserved for constructing self-supervised routing labels by comparing the outputs of the two generation strategies. The final evaluation set remains fully isolated and is used only for assessing the complete routed system.

This multi-stage splitting strategy supports both supervised model training and self-supervised routing induction while ensuring strict separation between training, routing, and evaluation phases.

4.4 Model Training and Generation Framework

Intent recognition is implemented using bert-base-uncased as a multi-class sequence classifier with one output class per intent. Input instructions are tokenized with truncation (maximum length 128 tokens). The model is trained using a learning rate of 2e-5 with gradient accumulation and step-based evaluation, and the best checkpoint is selected

according to validation performance (accuracy and macro-F1). In addition to intent prediction, the classifier provides a probability distribution over intents via softmax, which is later used to derive routing features.

To ensure realistic deployment conditions, all generation models condition on predicted intents rather than gold labels. The predicted intent is therefore added to each dataset split prior to response generation.

Two sequence-to-sequence response generators based on t5-small are trained using the Seq2SeqTrainer. The first model (T5-Simple) maps the user instruction directly to the reference response. The second model (T5-Complex) augments the input with the predicted intent in the format "intent: {intent_pred} | {instruction}", enabling explicit intent conditioning. The complex model uses longer input and output lengths to accommodate the augmented representation.

Both generators are trained with a learning rate of 3e-4, gradient accumulation, and step-based evaluation. Model selection is based on BLEU computed with sacreBLEU. While BLEU is not treated as an absolute quality metric, it provides a consistent automatic signal for comparing the relative performance of the two generation strategies.

4.5 Self-Supervised Routing and Threshold Calibration

A central contribution of this work is the construction of routing supervision without manual annotation. On the meta-development split, both generation models produce candidate responses for each instruction. For every example, per-example BLEU scores are computed between each generated response and the reference response. A routing label is then assigned according to which strategy achieves the higher BLEU score.

This procedure provides task-aligned weak supervision that directly answers the question: which generation strategy performs better for this query? Importantly, it avoids manual labeling of query complexity.

Routing decisions are learned using a logistic regression classifier trained on features derived from the intent classifier's output distribution. Each query is represented by four features: (1) the maximum softmax probability, reflecting confidence in the top intent; (2) the margin between the top two intent probabilities; (3) the entropy of the full intent distribution; and (4) the tokenized input length, serving as a proxy for query verbosity.

The router predicts the probability that a query should be processed by the intent-conditioned generation path. Rather than using a fixed decision threshold, threshold calibration is performed on the meta-development set. Thresholds are evaluated over a dense grid, optimizing macro-F1 while enforcing a practical constraint that the complex

routing rate remains between 5% and 95% to prevent degenerate solutions. The selected threshold reflects a trade-off between routing quality and computational balance.

4.6 Inference-Time Decision Process

At inference time, each user instruction is first processed by the intent classifier to obtain the predicted intent distribution. The four routing features are computed and passed to the trained logistic regression router, which estimates the probability of selecting the complex strategy. This probability is compared to the calibrated threshold to determine the generation path.

If the simple path is selected, the response is generated directly from the instruction. If the complex path is selected, the response is generated using the predicted intent in the same augmented format employed during training. This ensures consistency between training and deployment behavior.

5. Results and Discussion

5.1 Experimental Setup and Data Overview

Experiments are conducted on the Bitext Customer Support dataset (26,872 queries, 27 intents) using the four-way data split described in Section 4.3. The training and validation splits are used for model learning, the meta-development split for routing supervision and threshold calibration, and the evaluation split for final assessment. This separation ensures that routing decisions are learned without manual labels while preserving an unbiased test set.

5.2 Performance of Base Models

Intent Classification (BERT)

The BERT intent classifier achieves a Macro-F1 above 0.99 on the validation set, confirming that intent recognition is highly reliable on this dataset. However, high classification confidence does not necessarily imply low semantic complexity, motivating the need for dynamic routing.

Response Generation (T5)

The two T5 generation models achieve comparable BLEU scores overall. Qualitative inspection shows that the intent-conditioned model tends to produce more structured responses in procedurally complex cases, whereas the direct model performs adequately

on straightforward queries. These complementary behaviors justify the introduction of a routing mechanism rather than a single global strategy.

5.3 Routing Threshold Optimization and Trade-off Analysis

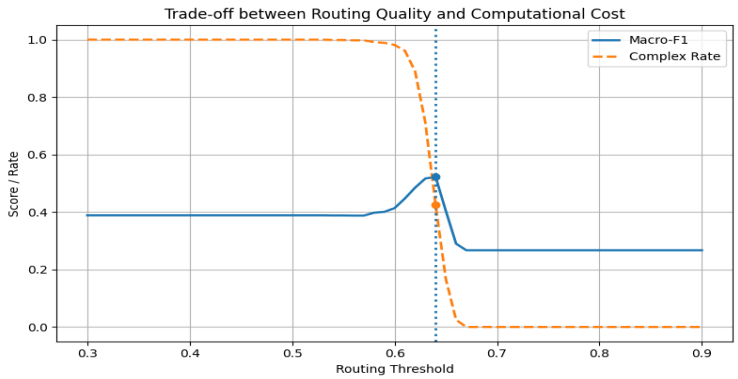

Figure 1: Trade-off between Routing Quality and Computational Cost. The non-monotonic behavior confirms that threshold calibration is necessary to avoid degenerate routing policies.

Figure 1 illustrates the trade-off between routing quality (Macro-F1) and computational cost (complex routing rate) as a function of the decision threshold. Routing performance exhibits a non-monotonic pattern. Low thresholds lead to nearly all queries being routed to the complex pipeline, resulting in degenerate behavior. As the threshold increases, a sharp transition region emerges where the router meaningfully discriminates between query types.

The optimal threshold (0.637) maximizes Macro-F1 while maintaining an approximately balanced routing rate (~50%). Beyond this region, performance degrades as the router increasingly defaults to the simple pipeline. These results demonstrate that routing quality depends on careful threshold calibration and that the trade-off between quality and efficiency is empirically grounded.

5.4 Routing Behavior Across Intents

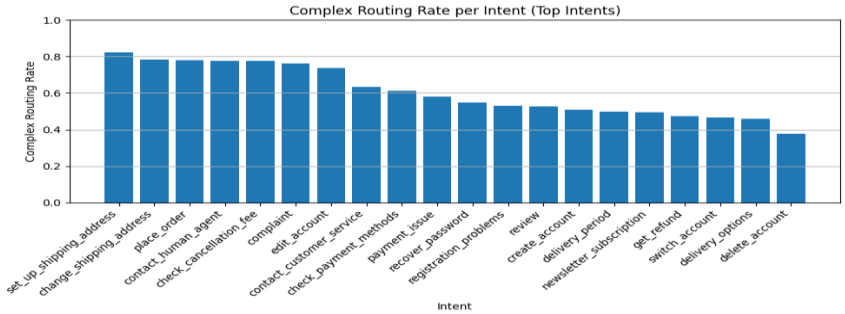

Figure 2: Complex Routing Rate per Intent. This figure reports the fraction of queries routed to the complex pipeline for the most frequent intents in the dataset. Each bar corresponds to a single intent, sorted by complex routing rate.

Figure 2 reveals substantial variation in routing behavior across intents. Intents such as *set_up_shipping_address*, *complaint*, and *contact_human_agent* are predominantly routed to the complex pipeline, reflecting their higher semantic or procedural complexity. In contrast, intents like *get_invoice* or *check_invoice* are mostly handled by the simple pipeline, suggesting that they are well-served by direct generation.

Crucially, routing decisions are not trivially determined by intent labels alone. Several intents exhibit intermediate routing rates, indicating that query-level features, rather than intent identity, drive routing decisions. This supports the design choice of performing routing dynamically at inference time.

5.5 Feature Analysis and Robustness

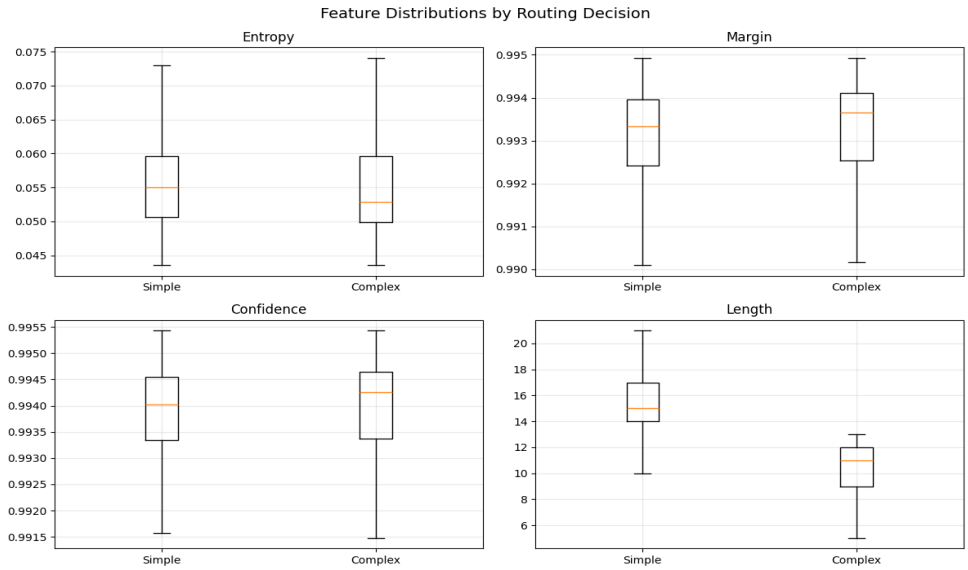

Figure 3: Feature Distributions by Routing Decision. This figure compares the distributions of four routing features—entropy, margin, confidence, and input length—between queries routed to the simple and complex pipelines. Each boxplot shows the median, interquartile range, and overall dispersion.

As illustrated in Figure 3, feature distributions reveal substantial overlap between simple and complex routing decisions. Confidence, margin, and entropy do not independently explain routing outcomes, suggesting that semantic complexity cannot be reduced to classification uncertainty alone. Input length shows a clearer separation, although routing ultimately emerges from the interaction of multiple weak signals.

A robustness analysis using minimal linguistic perturbations shows that routing decisions remain stable for approximately 95% of tested queries. Occasional flips are typically triggered by subtle semantic variations rather than superficial changes, indicating that the router balances sensitivity and stability.

5.6 Discussion and Implications

Overall, the results demonstrate that dynamic query routing is feasible and beneficial in customer service settings. The router learns a non-trivial decision boundary that balances response quality and computational cost without manual supervision. Routing behavior aligns with intuitive notions of complexity while remaining flexible at the query level.

These findings support the central claim of this work: intent recognition and end-to-end generation are complementary, and selectively combining them through learned routing yields a more adaptive conversational system.

6. Conclusion

This work addressed the challenge of handling diverse user queries in customer-service conversational systems, where a single fixed generation strategy may be suboptimal. We proposed a dynamic hybrid framework that integrates intent recognition and end-to-end generation within a unified architecture, enabling per-query strategy selection through a learned routing mechanism.

By constructing routing supervision through self-supervised comparison of generation outputs, the proposed approach avoids manual labeling while learning a non-trivial decision boundary. Experimental results demonstrate that dynamic routing achieves a balanced trade-off between response quality and computational cost, adapts across intents, and remains stable under minor linguistic perturbations.

Overall, the findings show that explicit intent recognition remains valuable when applied selectively. Rather than replacing traditional pipelines, modern generative systems benefit from dynamically combining structured intent signals with flexible end-to-end modeling. This work demonstrates a practical and principled approach to reconciling control and adaptability in contemporary conversational AI.

7. Future Work

Future work could extend this study in several complementary directions. The routing mechanism may benefit from incorporating additional signals derived directly from the generative models, such as decoding uncertainty or token-level statistics, which could help refine routing decisions in borderline or ambiguous cases. While the current evaluation

relies on a controlled customer service dataset, validating the proposed framework on real-world conversational data and across multiple application domains would provide stronger evidence of its robustness and generalizability. Moreover, the use of a fixed global routing threshold could be reconsidered, as adaptive or intent-specific thresholding strategies may allow the system to better align routing behavior with operational constraints such as latency, reliability, or computational cost. Finally, future research could explore joint optimization approaches in which the router and downstream generative models are trained in a more tightly coupled manner, potentially improving overall coordination and end-to-end performance.

Code Availability

The complete implementation of the dynamic hybrid chatbot framework, including model training, routing, and evaluation scripts, is publicly available at:

https://github.com/anaellez/Dynamic-Hybrid-Chatbots-Learning-When-to-Rely-on-Intent-Recognition-or-Direct-Generation/tree/main

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